

How a school swarms

Why fish swarm, circle and swim together

Lengths are in **body lengths** (BL; a fish is 3.5 in). The tank is 13.4 × 3.4 × 4.7 BL inside. A **school** is polarised (p_group near 1), a **swarm** is not: moving the zone width Δr_o switches between them.

A · Four collective states from one rule, simulated here (N = 100, the paper's r_r, α, θ, s, σ)

A · swarm



top view

B · torus



spin-axis view

C · dynamic parallel



side view

D · highly parallel



side view

Δr_o = 0, Δr_a = 15
p_group 0.16 · m_group 0.10

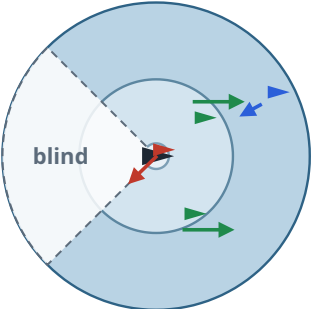
Δr_o = 1, Δr_a = 12
p_group 0.09 · m_group 0.90

Δr_o = 3, Δr_a = 10
p_group 0.90 · m_group 0.03

Δr_o = 10, Δr_a = 10
p_group 1.00 · m_group 0.02

ours · N = 100 simulated here; the zone widths are ours, the paper prints none

B · The model: three zones and a blind volume



repulsion zone, r_r = 1
anyone here: turn away, nothing else counts

orientation zone, width Δr_o
those here: match their heading

attraction zone, width Δr_a
those here: turn toward them

blind volume, 90° behind
α = 270°: cannot see here

Drawn at the school setting: Δr_o = 5, Δr_a = 6

verified · zones, α, r_r: Couzin 2002

ours · widths drawn

C · The map: p_group over the two zone widths

attraction width Δr_a

orientation width Δr_o (BL)

15	S	T	?	D	D	P	P	P	P	P	P	P	P	P
12	S	T	?	D	D	P	P	P	P	P	P	P	P	P
10	S	?	D	D	D	P	P	P	P	P	P	P	P	P
8	S	F	D	D	D	P	P	P	P	P	P	P	P	P
6	F	F	D	D	D	P	P	P	P	P	P	P	P	P
4	F	F	F	D	D	P	P	P	P	P	P	P	P	P
3	F	F	F	F	D	P	P	P	P	P	P	P	P	P
2	F	F	F	F	D	P	P	P	P	P	P	P	P	P
1	F	F	F	F	F	P	P	P	P	P	P	P	P	P
0	F	F	F	F	F	F	P	P	P	P	P	P	P	P
	0	1	2	3	4	6	8	10	12	15				

p_group (alignment)

< 0.25

< 0.6

< 0.9

≥ 0.9

S swarm: low p, low m
T torus: low p, high m
D dynamic parallel: high p
P highly parallel: p ≈ 1
F fragments (more than half)
? seeds disagree: no one state

ours · N = 100, 3 replicates, 700 steps (paper: 30)

D · The rule in eight lines, and the Forth word that does each

1 See the others, except behind

2 Repulsion if anyone is within r_r

3 Else orientation and attraction

4 Nobody in any zone: keep the heading

5 Turn toward d by at most θτ, move s τ

6 The leader counts w times in sums 3

7 Walls repel like a neighbour, and are a hard limit

8 A dart startles the followers nearby

u_ij · v_i ≥ cos(α/2)
d = -Σ u_ij (nothing else counts)
d_o = Σ v_j, d_a = Σ u_ij; d = (d_o + d_a)/2
d = v_i
v ← rotate(v, d, θτ); c ← c + s τ v
Σ w_j v_j, Σ w_j u_ij
within 0.6 BL: one more repulsion; past the box: put back, heading reflected
heading flips away at once; 2.5 × speed for 0.6 s

sch-pair

sch-pair

sch-pair, sch-want

sch-want

sch-turn, sch-follow

sch-pair (weight)

sch-wall, clamp-axis

sch-startle-all

The paper's random error (σ = 0.05 rad) is left out of the Forth, and the game adds none. Our units differ from the paper's, on purpose: at its s = 3 BL/s and θ = 40°/s a 90° turn takes 6.75 BL of swimming (radius 4.3 BL), wider than the tank is deep (3.4 BL), and the tank runs below could not keep the fish in. Ours: s = 2 BL/s, θ = 120°/s, 1.50 BL, radius 0.95 BL.

verified · rules 1–5 and the paper's s, θ: Couzin 2002, eqns 1–3 and Fig. 3

ours · rules 6–8, our s, θ, the 0.6 BL wall zone, the 0.6 s startle

F · The Forth core, school.fth: the four rule words, and what they cost

sch-pair	168 B · 24 lines	sch-want	88 B · 11 lines	sch-turn	104 B · 12 lines	sch-follow	131 B · 17 lines
one neighbour: blind volume, then repulsion, orientation or attraction : sch-pair (j --) dup MB_YOU sc! sch-stride * sch-base + dup read-mailbox MB_PERP sc@ - MB_U sc! dup 1 + read-mailbox MB_PERP 1 + sc@ - MB_U 1 + sc! 2 + read-mailbox MB_PERP 2 + sc@ - MB_U 2 + sc! MB_U vlen2 dup 0.000001 > over MB_R2 sc@ < & if sch-sqrt dup MB_DIST sc! 1 swap / MB_U vscale heading-dot-u MB_COSB par@ >= if r-rep inside if -1 MB_SUM_R acc+u MB_N_R count+ else r-ori inside if you-weight MB_SUM_0 acc+v MB_N_0 count+ else you-weight MB_SUM_A acc+u MB_N_A count+ then then then else drop then ;	the wanted direction from the three sums, or the old heading : sch-want MB_WANT vzero MB_N_R seen? if MB_SUM_R add-wanted else MB_N_0 seen? if MB_SUM_0 add-wanted then MB_N_A seen? if MB_SUM_A add-wanted then then MB_WANT vnorm 0.000001 > not if MB_WANT heading>sc then ;	turn toward it by at most θτ, keep the heading unit length : sch-turn MB_VX me sch@ MB_WANT sc@ * MB_VY me sch@ MB_WANT 1 + sc@ * + MB_VZ me sch@ MB_WANT 2 + sc@ * + dup MB_COST par@ >= if drop else perp-all MB_PERP vnorm 0.000001 > not if perp-fallback then turn-all MB_WANT vnorm drop then ;	one follower, start to finish: scan, walls, startle, turn, : sch-follow dup MB_ME sc! clear-sums sch-scan-cache sch-n 0 do i over <> if i sch-pair then loop drop sch-walls sch-want sch-startle? if sch-startle-away sch-startle-tick then sch-turn MB_WANT sc@ MB_NVX me sch! MB_WANT 1 + sc@ MB_NVY me sch! MB_WANT 2 + sc@ MB_NVZ me sch! 0 advance 1 advance 2 advance clamp-all ;				

3,324 B

of the 64 KB dictionary (5.1 %), 210 code lines; 7.3 ms a step on the Chromecast (at 3,324 B); 1e-03 off couzin.py

7.0 → 11.3 ms

one tick, 11 fish, Chromecast HD (an earlier 2,936 B version): the bare interpreter, then in the engine

4.2 → 0.28 μs

per mailbox call: three debug streams the engine ran on every call, found by this timing; 20 fps before

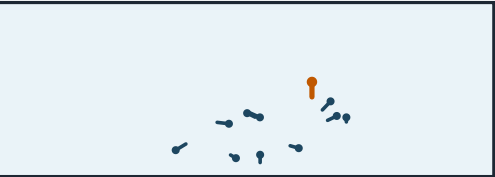
59.9 fps

the real level with ten followers, Chromecast HD, unprofiled; p90 33.4 ms, Director 6.4 ms (profiled)

ours · our measurements: the bare interpreter, and a bench level (the Director runs school.fth every tick) on the real Chromecast, release build

G · The Forth, running in the tank: swarm, torus, school, and a startle

swarm setting (0, 10)



leader weight w
w = 1 2.9 BL · -0.03 · 0.30 · yes
w = 3 2.5 BL · +0.03 · 0.41 · yes
w = 6 2.3 BL · +0.11 · 0.45 · yes

torus setting (1, 10)



leader weight w
w = 1 2.8 BL · +0.12 · 0.59 · yes
w = 3 2.3 BL · +0.25 · 0.59 · yes
w = 6 2.1 BL · +0.20 · 0.57 · yes

school setting (5, 6)



leader weight w
w = 1 3.8 BL · +0.03 · 0.86 · yes
w = 3 3.8 BL · -0.00 · 0.85 · yes
w = 6 3.1 BL · +0.09 · 0.78 · yes

Startle: distance to the leader



swarm

torus

school

—6 s to +12 s; top = 6 BL; one seed

In the real level, on the engine: resting p_group 0.39, swimming 0.77, followers aligned with the leader +0.40; a dart: their distance to it 1.55 → 4.17 BL in 1 s, back to 2.4 in 4 s.

ours · the Forth core in the tank's real box (w = 3, seed 1 drawn; the table is the mean of 3 seeds; "no" = a fish strayed more than 0.5 BL out)

Sources · checked sources; model results and game choices distinguished

S1 Couzin et al. 2002, J. Theor. Biol.

verified

 zone model, states, metrics, Fig. 3

S2 Couzin et al. 2005, Nature

verified

 simple model: informed minority guides a group

S3 Pitcher 1998, Shoaling and Schooling

verified

 schooling: coordinated swimming within a shoal

S4 Madduppa et al. 2014, PLOS ONE

verified

 adults in anemone-associated social groups

S5 the aquarium level and this repository

ours

 tank, fish, zForth, our runs

How to read the chips

verified

 claim checked in the source

ours

 our own choice, maths or measurement

Game choice: schooling. Real ocellaris: anemone-associated groups. Sweep too coarse for hysteresis; lead untuned. Data 2026-10-01; sources checked 2026-10-06.